AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 17 · STUDENT LESSON

Noise and Outliers

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Calculate two summaries. Investigate the unusual record.

Distinguish a mathematical what-if, a verified repair, a real rare event and missing evidence.

Keep one habit: Five durations → mean and median → odd/even/empty procedures → source-backed audit.

MAIN LESSON · SECTION 1

Does an unusual value mean the record is wrong?

> Calculate two summaries and investigate the source before changing evidence.

Your learning goals

I can calculate **mean** and **median**.

I can **investigate** an **unusual value** before deleting it.

I can explain how a summary can hide **variation**.

Recall a total before dividing

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- What is the total of 2, 3 and 4?

Your end goal

By the end, I can calculate **mean** and **median** and **investigate** an **unusual value** before changing it.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Over**fitting** Intuition. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from **Training** and Testing. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 17: **Noise** and **Outlier**s

Mistakes and **unusual value**s should be **investigate**d, not automatically erased.



THINKER MISSION

Decide whether four odd records are errors or real cases.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week’s end goal

MAIN LESSON · SECTION 2

Keep all five durations while asking why one is long

> The observed values alone do not justify deletion.

The complete fictional duration list

Five fictional activity durations are 2, 3, 3, 4 and 18 minutes. Most are near 3, but one is much longer. Different summaries reveal different aspects.

Keep all values while investigating whether 18 is a genuine long activity or a recording error.

Separate unusual observations from confirmed errors

Unusual means worth investigating in the context of these records; this lesson does not define a universal numerical outlier cutoff. A long genuine task is different from a transcription mistake or a duplicated row. Neither the mean nor the median can identify the cause by itself. Keep units, event identity and the original source visible.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use **noise** and **outlier**s to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

Write or draw your reasoning and evidence.

My **prediction** and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

MAIN LESSON · SECTION 3

Trace mean and odd-count median with complete Python

Explain the count and sorted position before trusting the output.

Predict and run the five-value script

```
values = [2, 3, 3, 4, 18]
ordered = sorted(values)
mean = sum(values) / len(values)
median = ordered[len(ordered) // 2]
print(mean)
print(median)
```

Read the syntax and its limited contract

sum adds all observations, including repeated values; len counts observations, not distinct numbers. sorted makes a separate ordered list. With five entries, len//2 is2, so zero-based position2 is the third value. That shortcut selects the median only for a nonempty odd count. The original list is preserved for checking against the source.

The script prints 6.0 and 3. sorted returns a sorted list without changing the original. This **median** expression works for a nonempty odd-length list only. For four sorted values, average the second and third.

Do not use this odd-length shortcut for every dataset.

Compare the two summary questions

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
Sorted data	2, 3, 3, 4, 18	Middle is 3
Mean	30 divided by 5	6.0
Typical middle	Third of five values	3

Read each row from left to right and connect the evidence to the stated rule.

The **mean** shares the total equally: 2 + 3 + 3 + 4 + 18 is 30, and 30 divided by five is 6. The **median** asks for the middle value after sorting: the third value is 3.

The long duration 18 pulls the **mean** upwards but leaves the middle value at 3. A difference between the summaries is a clue to inspect the data, not permission to remove the inconvenient value.

Point to a deciding calculation

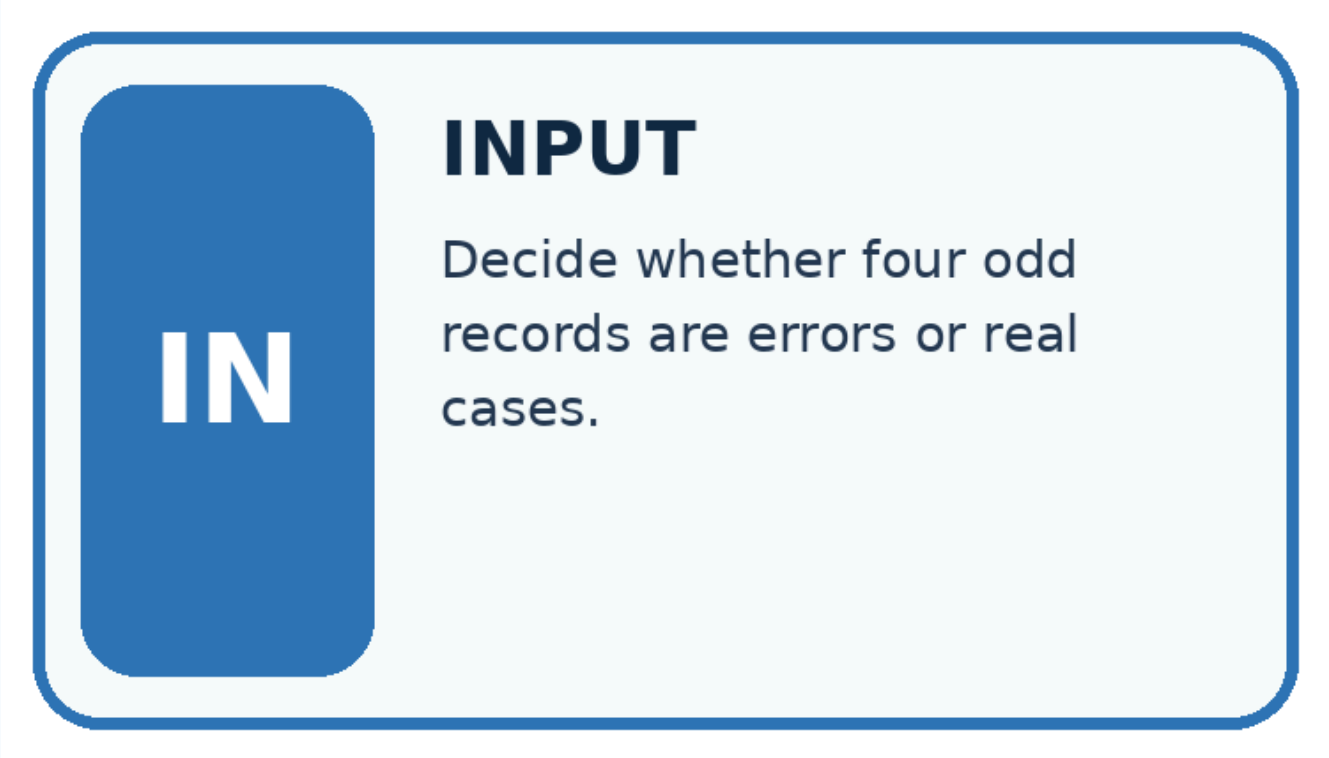
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

VISUAL **MECHANISM**

Input → **Process** → **Output**

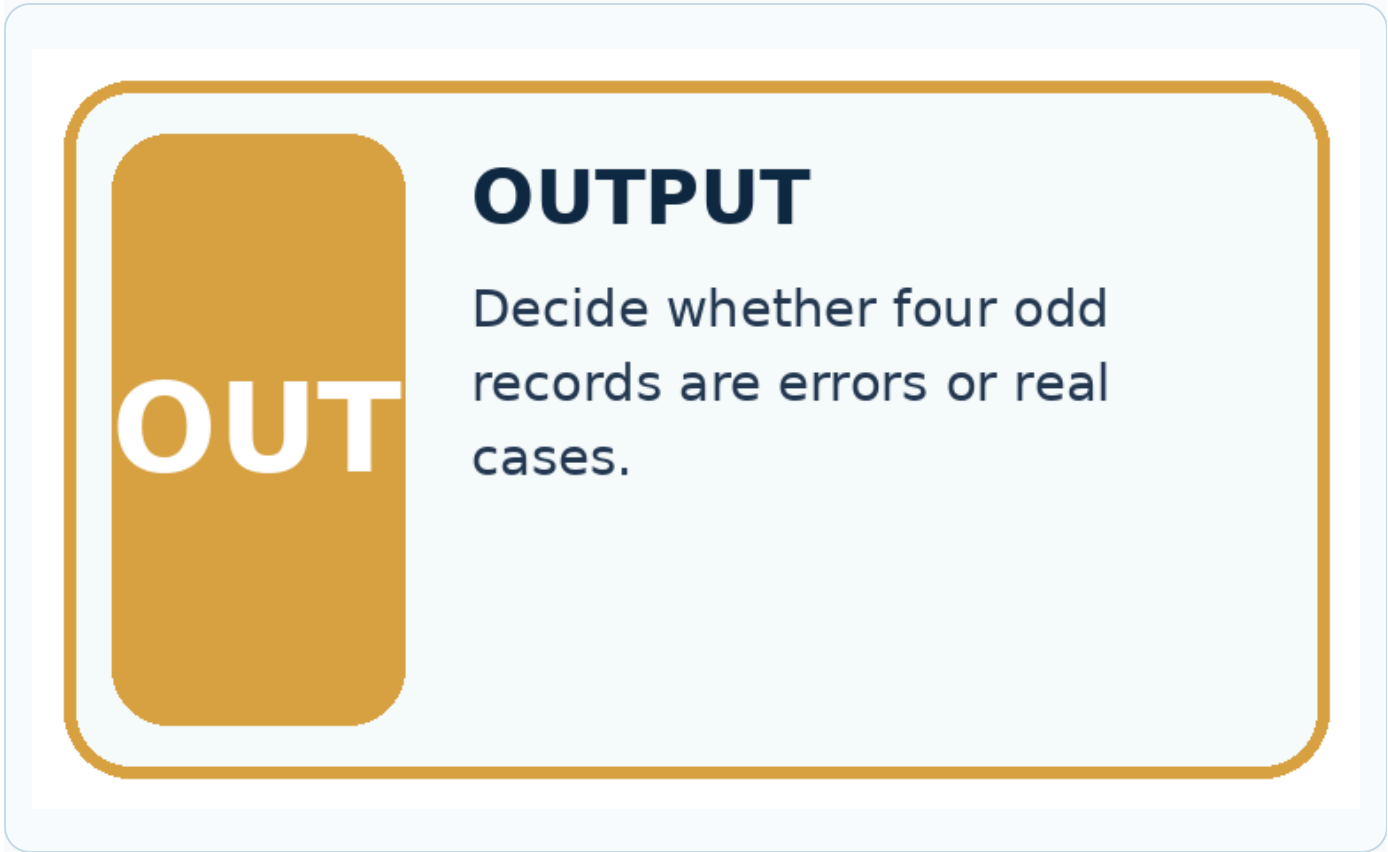
Trace the information. At each stage, ask what changed and how you could test it.



↓


PROCESS
Mistakes and unusual values should be investigated, not automatically erased.

↓


OUT
OUTPUT
Decide whether four odd records are errors or real cases.

Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

Use two summaries for two questions

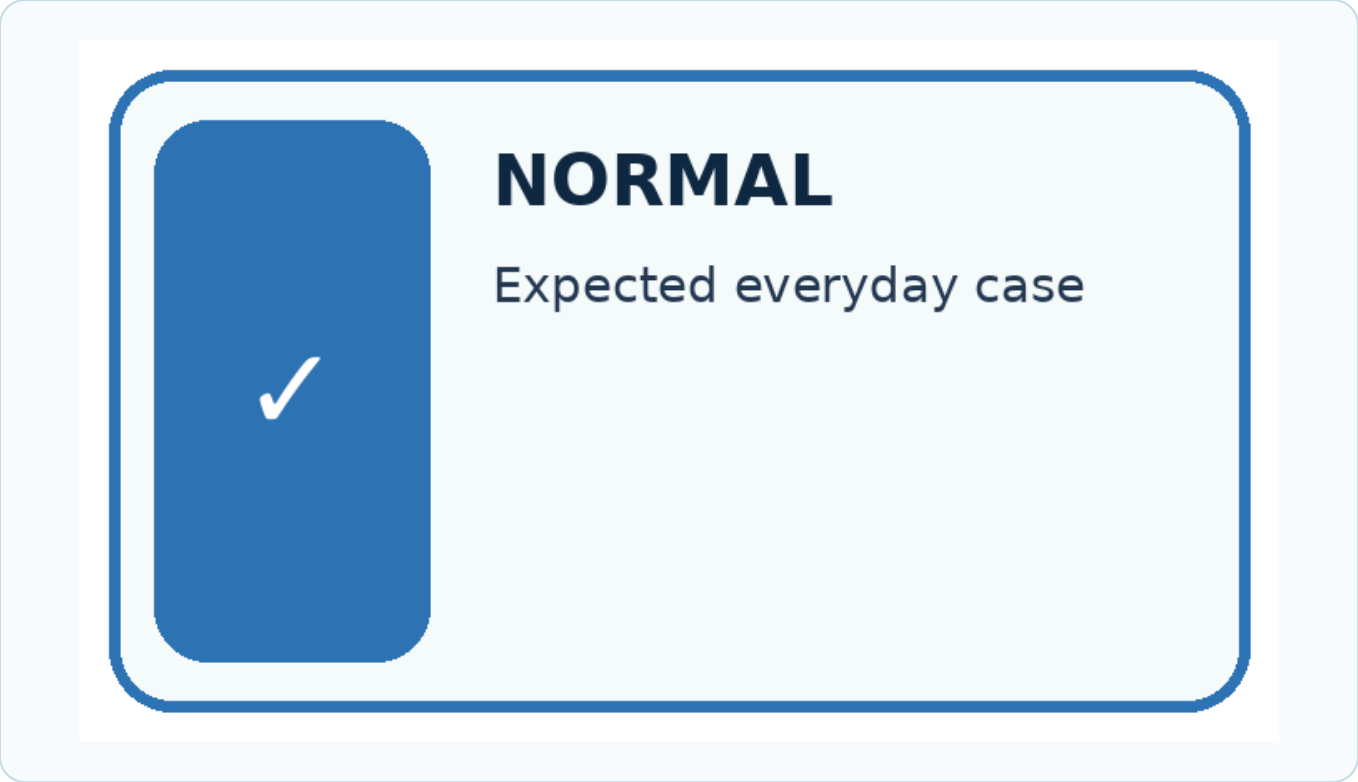
Connect the picture and the explanation

MAIN LESSON · SECTION 4

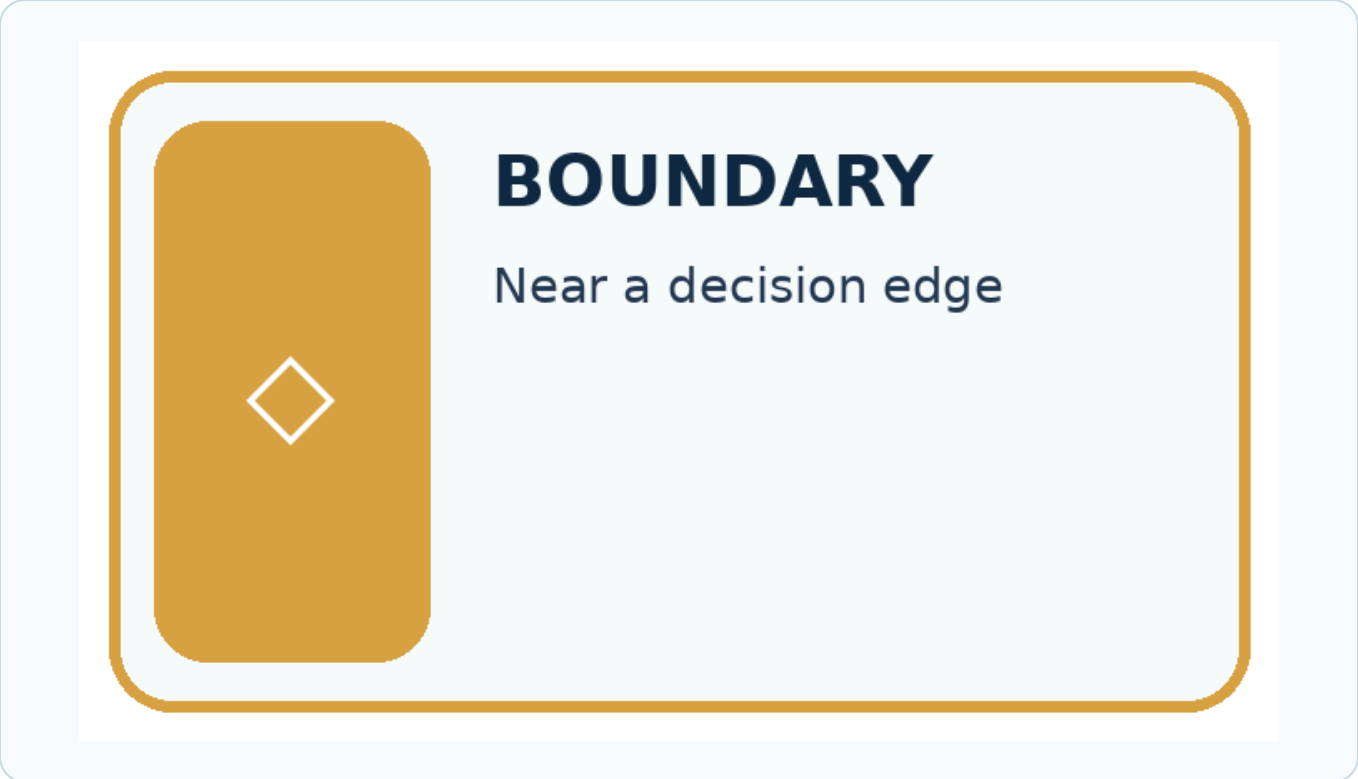
Compare a typical middle and a source-backed repair

> Arithmetic sensitivity and permission to change data are different questions.

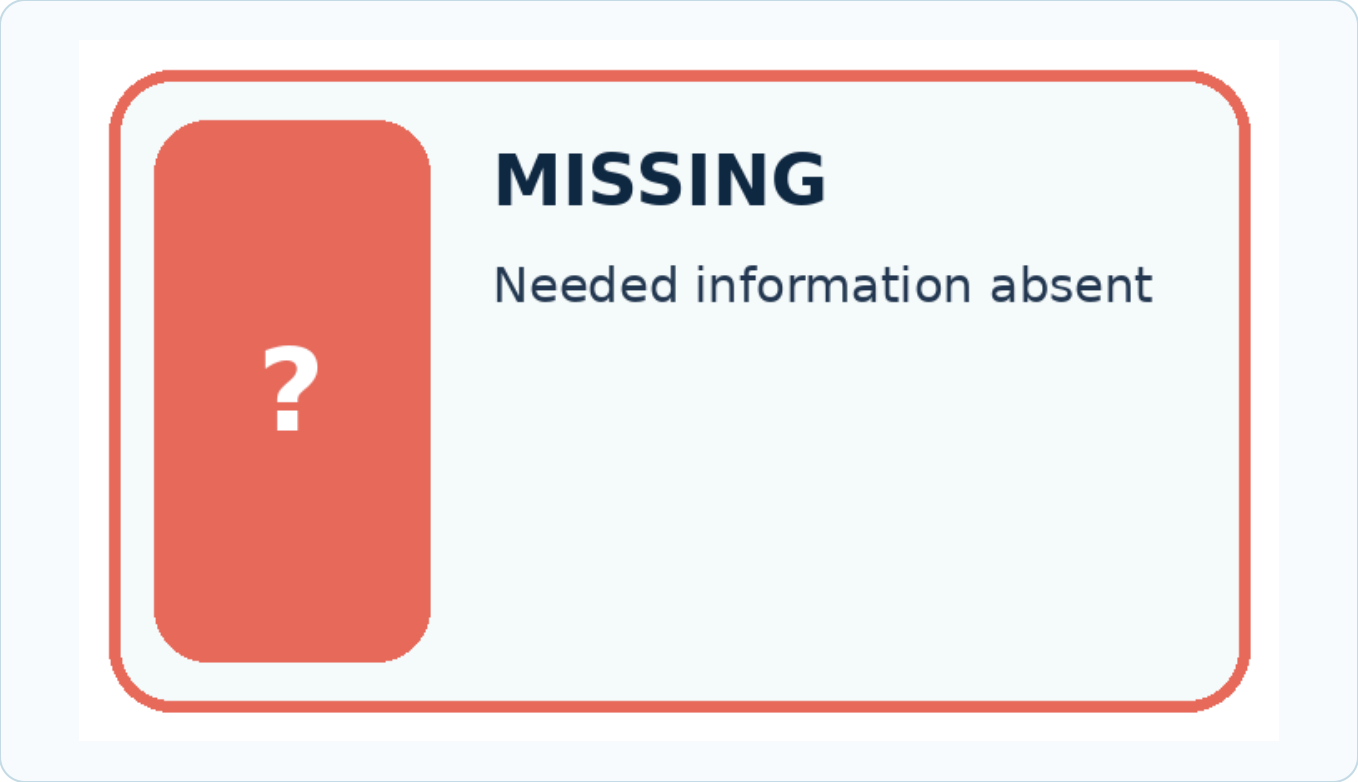
Four pictures with a purpose



Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: An authorised log confirms 18 was a transcription error for 3.

▼ Check after predicting

Prediction: Corrected **mean** is 3 minutes and **median** is 3.

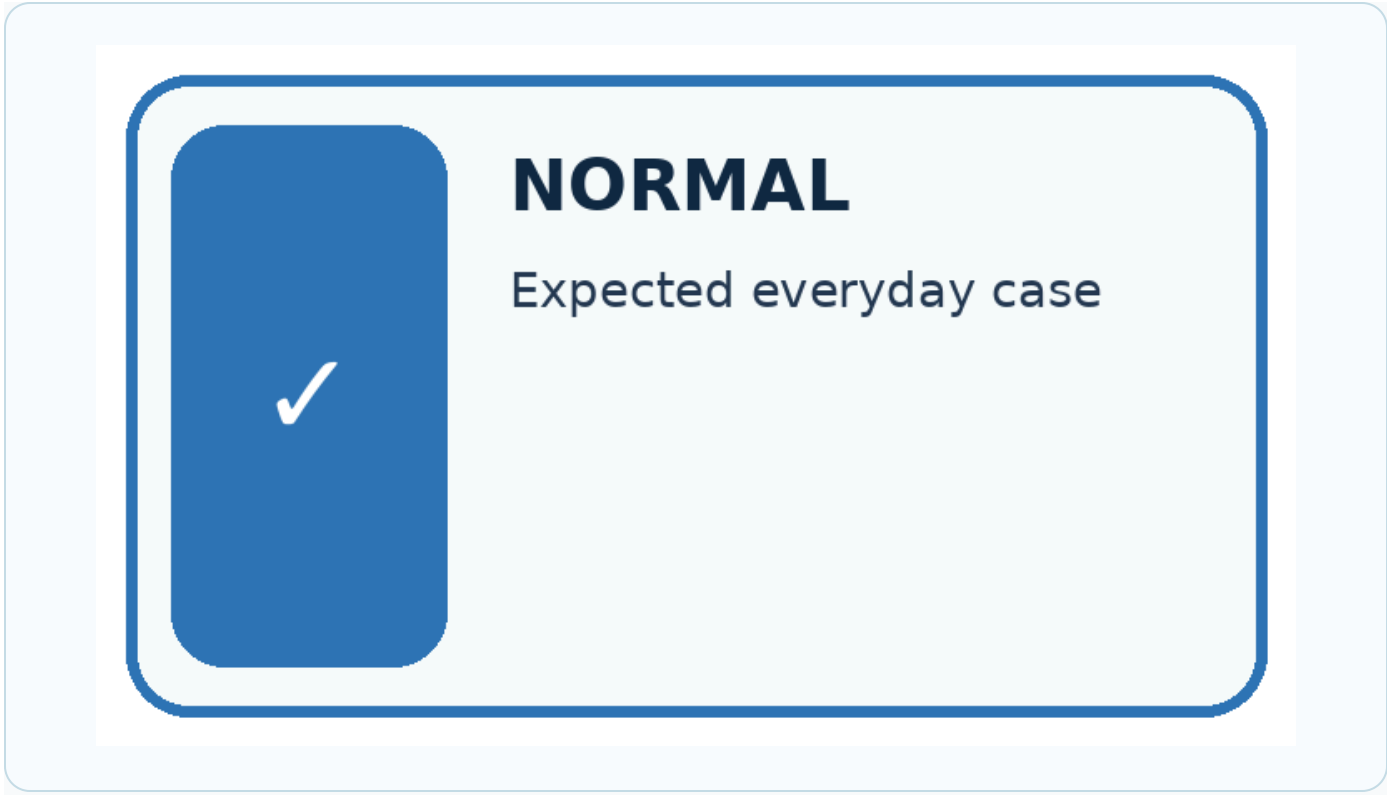
Reason: Values become 2,3,3,4,3 with sum 15; sorted middle remains 3.

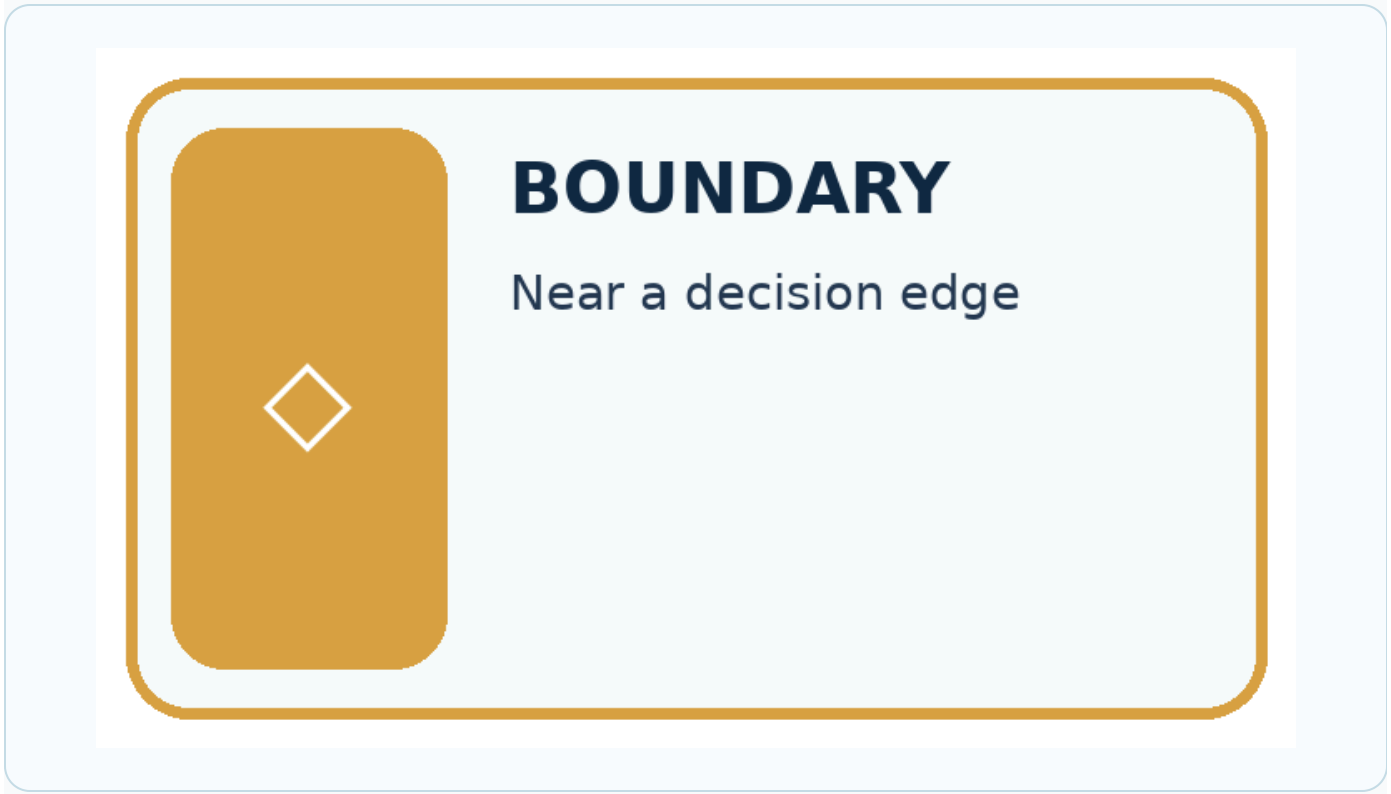
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

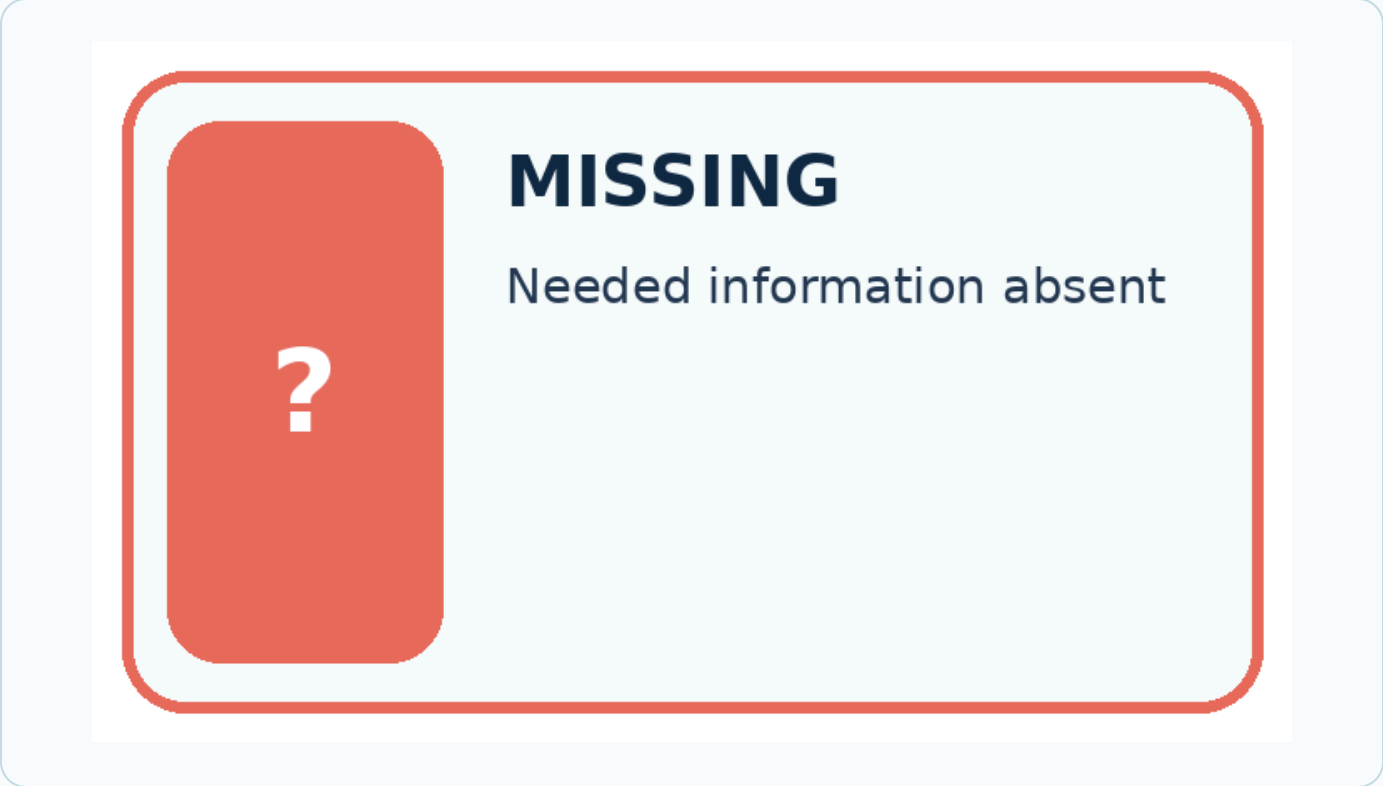
Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Calculate a what-if change, then examine an authorised correction

> The same baseline values can support different explicitly labelled comparisons.

Separate the two change scenarios

Replacing18 with8 below is a hypothetical sensitivity calculation, not a claim that the source was wrong. The pictured authorised-log scenario instead supplies evidence for18→3. Keep those two changed lists separate. Both use minutes; the picture names the fictional tasks as book processing.

Try18→8 as a what-if calculation

Replace 18 with 8. Calculate the new mean and median.

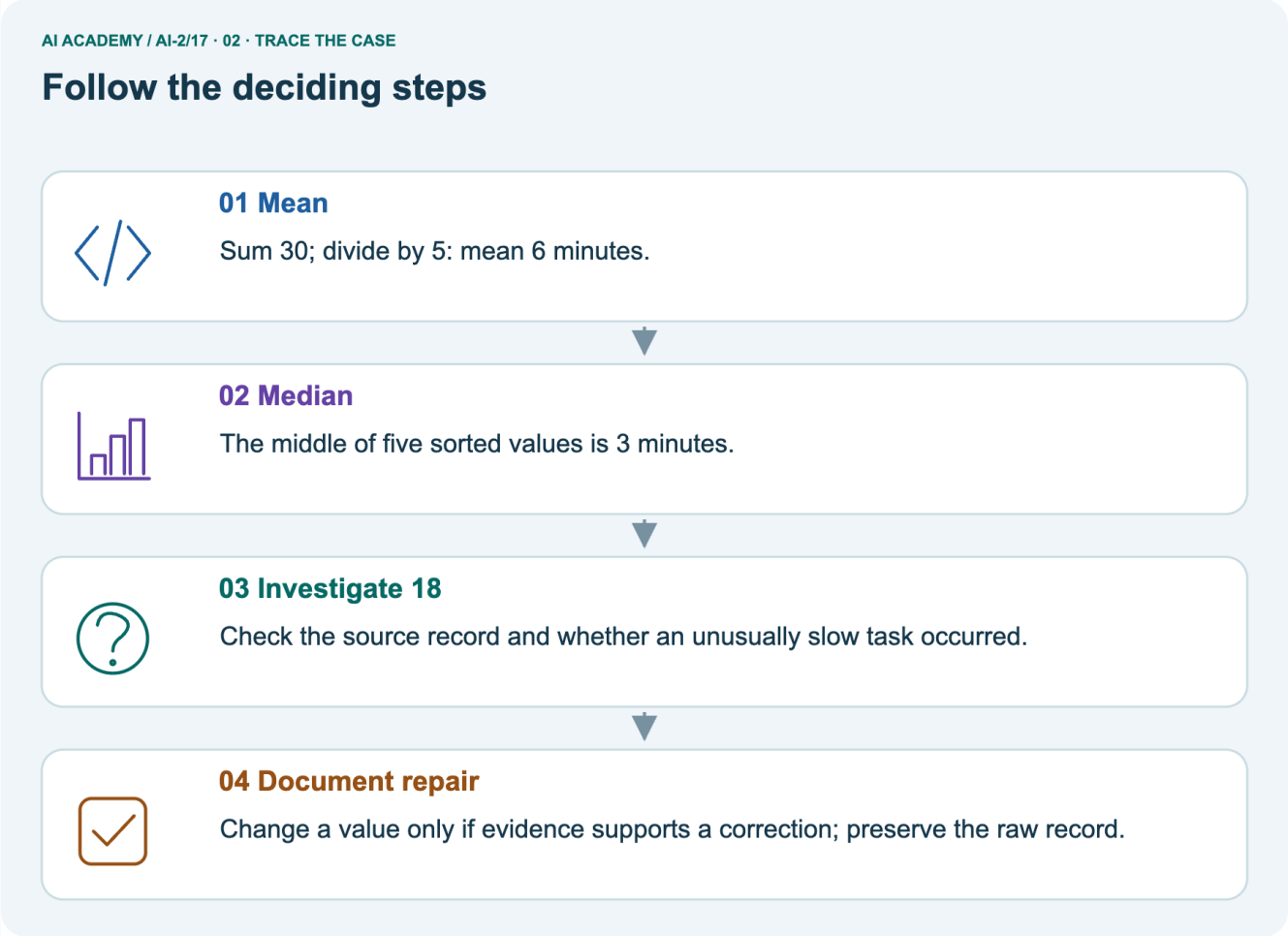
Read the pictured baseline and source status

A fictional library records five book-**process**ing times in minutes.

Sorted times: 2,3,3,4,18.

All are recorded measurements; no **evidence** yet shows 18 is a mistake.

Trace mean, median and the need to investigate



Follow the labelled picture with the matching explanation. Open the vector diagram

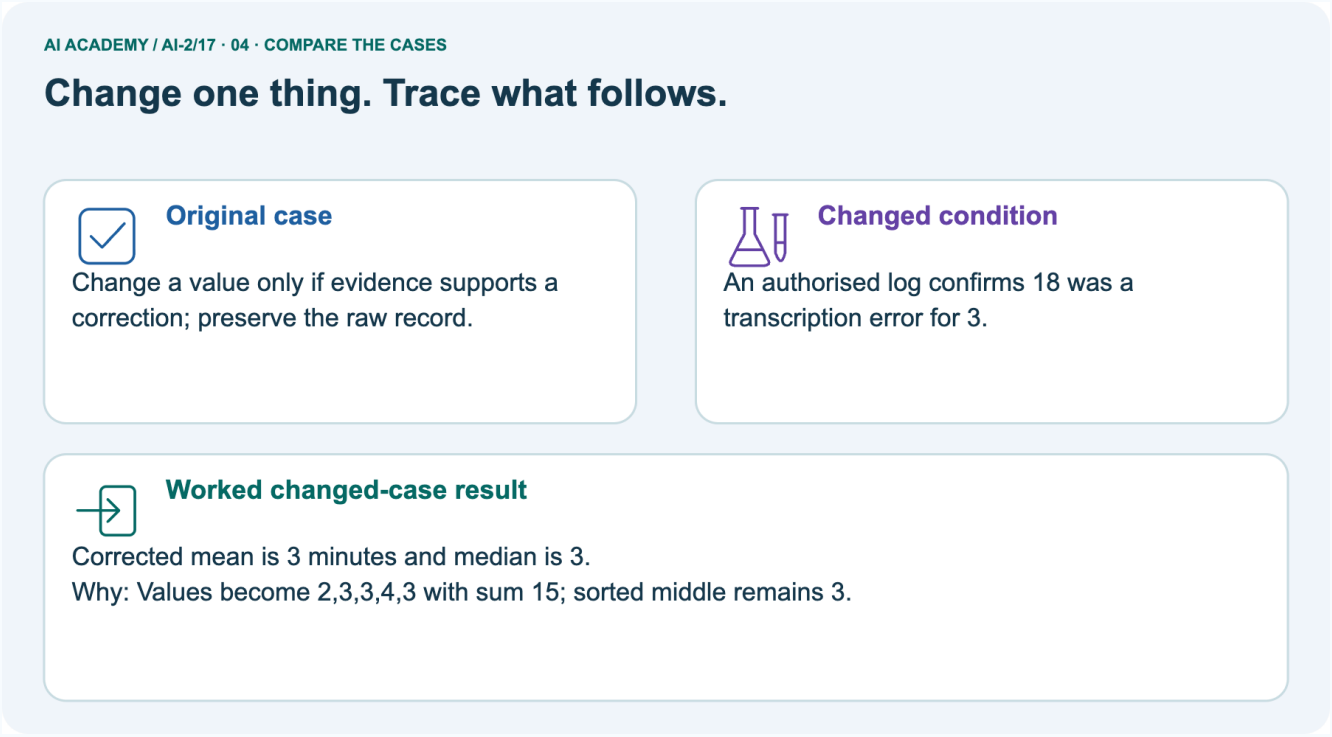
The **mean** uses every value and is pulled upward by 18. The **median** identifies the middle recorded time and stays at 3. Neither statistic declares the unusual **observation** invalid.

Deleting it simply because it changes the **mean** would hide potentially real slow work and change the question being answered.

Predict the authorised18→3 correction

- Change one thing: An authorised log confirms 18 was a transcription error for 3. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain the middle task and the limits

- Read the labelled picture: If 18 was genuine, which summary better describes the middle observed task, and what extra fact should be shown?
- Five task times do not estimate all library **process**ing times reliably or prove which summary suits every decision.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 17 • WORKED EVIDENCE

1. Most books: 40–300 pages; one record: 3,200 pages.
2. Check the source: extra zero, box set, or real long work?
3. Repair confirmed error; keep real rare case and document it.
4. Retest to learn whether the system handles legitimate extremes.

1. Most books: 40–300 pages; one record: 3,200 pages.

2. Check the source: extra zero, box set, or real long work?

3. Repair confirmed error; keep real rare case and document it.

4. Retest to learn whether the system handles legitimate extremes.

Recalculate, retrace, or restate the reasoning in your own representation:

> **Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

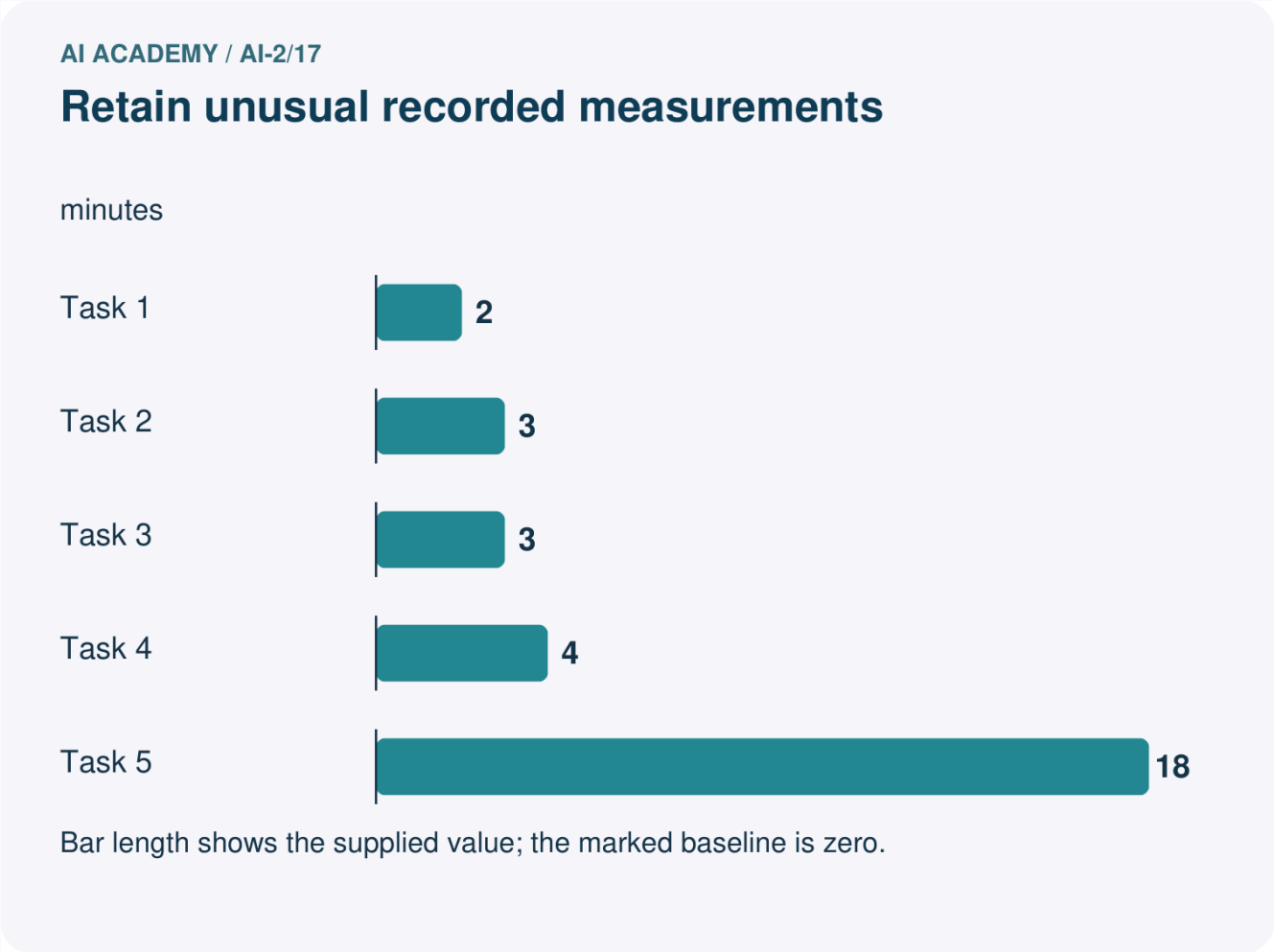
A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Noise and Outliers

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/17

Noise and Outliers

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Mean

Sum 30; divide by 5: mean 6 minutes.



2. Median

The middle of five sorted values is 3 minutes.



3. Investigate 18

Check the source record and whether an unusually slow task occurred.



4. Document repair

Change a value only if evidence supports a correction; preserve the raw record.

CHANGE ONE THING

An authorised log confirms 18 was a transcription error for 3.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Mean	Sum 30; divide by 5: mean 6 minutes.
2. Median	The middle of five sorted values is 3 minutes.
3. Investigate 18	Check the source record and whether an unusually slow task occurred.
4. Document repair	Change a value only if evidence supports a correction; preserve the raw record.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Do not hide an inconvenient observation

> A missing summary is different from a measured zero.

Preserve evidence and document sensitivity comparisons

Removing 18 merely because it raises the **mean** would hide **evidence**. Check collection notes, keep an audit trail and compare summaries with and without it only with a clear explanation.

Challenge a zero mean for an empty dataset

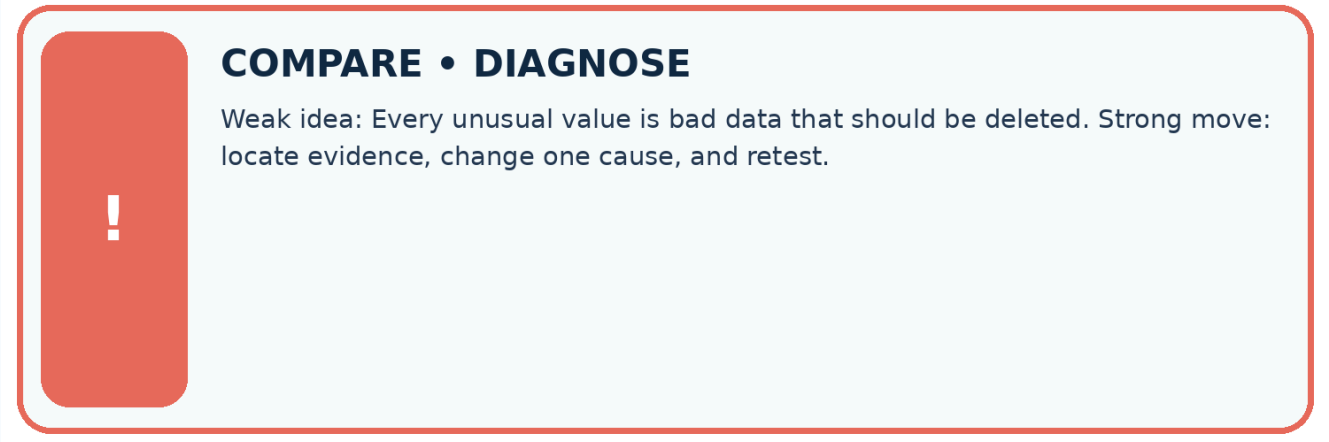
> **A dataset is empty. Is reporting a mean of 0 justified?**

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.



Evidence record	Expected	Actual	Likely cause / next test
normal case	works		

Evidence record	Expected	Actual	Likely cause / next test
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 8

Sort first and handle even and empty lists explicitly

> Do not reuse an odd-count shortcut as a universal median procedure.

Compare unsorted position with sorted median

Taking the middle item of an unsorted list can give the wrong **median**. For [18, 2, 4, 3, 3], the middle item is 4, but the sorted **median** is 3. Sort first. For an empty list, there is no **mean** or **median** under this lesson's rules.

Try both workbook medians before using the extension

Use workbook W2 for the connected example "**Mean median** and **unusual value**s". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Read the complete three-case extension

```
def summaries(values):
    if not values:
        return None, None
    ordered = sorted(values)
    middle = len(ordered) // 2
    if len(ordered) % 2 == 1:
        median = ordered[middle]
    else:
        median = (ordered[middle - 1] + ordered[middle]) / 2
    mean = sum(values) / len(values)
    return mean, median

for values in [[2, 3, 3, 4, 18], [8, 2, 4, 6], []]:
    print(summaries(values))
```

Explain empty, odd and even branches

This extended function assumes finite numeric observations in consistent units. %2 gives remainder1 for an odd count. For an even count, average positions middle-1 and middle in the sorted list. The empty branch returns None,None to mean unavailable, not measured zero. The outputs are (6.0,3), (5.0,5.0) and (None,None). The function does not decide whether an unusual value is a valid measurement.

Challenge successful execution proves the conclusion

- Use the worked case for **Mean median** and **unusual value**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Return to the pictured evidence question

If 18 was genuine, which summary better describes the middle observed task, and what extra fact should be shown?

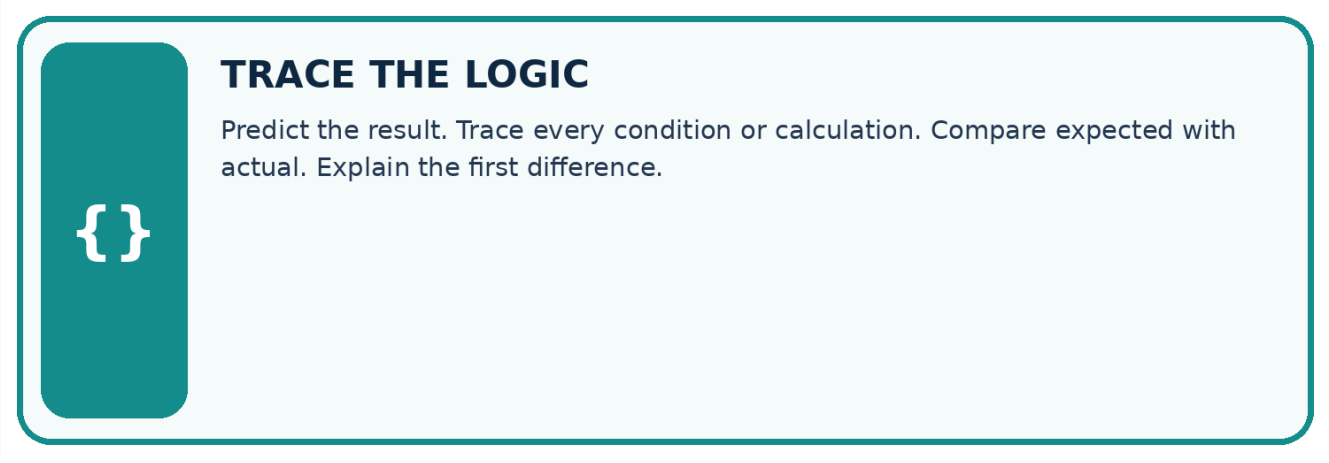
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

Practise with less help: Compute two medians

Check your reasoning

Explain the picture

> **My explanation using a pictured step and the given facts**

MAIN LESSON · SECTION 9

Audit four supplied records using explicit source evidence

> Keep real long tasks, confirmed errors, duplicate copies and missing measurements distinct.

The complete audit-card evidence

Record / event	Recorded duration	Source evidence supplied
R1 / E1	18 minutes	Authorised original log clearly records3 minutes
R2 / E2	25 minutes	Approved log verifies a genuine25-minute task
R3 / E2	25 minutes	Export log confirms this is a duplicate copy of R2, not another event
R4 / E3	blank	Original duration is unreadable and no replacement measurement is available

Predict, investigate and retain provenance

1. These four audit records form a separate investigation, not replacements for the five-value baseline list. One analysed observation should represent one distinct processing event, in minutes. For each record predict keep, correct, exclude duplicate from the analysis, or ask for evidence. Cite the exact supplied source fact. Preserve the raw record even if its analysis copy is corrected or excluded. For an unresolved duration, report unavailable or review; do not make it zero. Record source, old value, new value or action, reason and what remains unknown. No actual source files are being changed; label the result a paper audit.

Keep the evidence record and explain your hardest decision

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

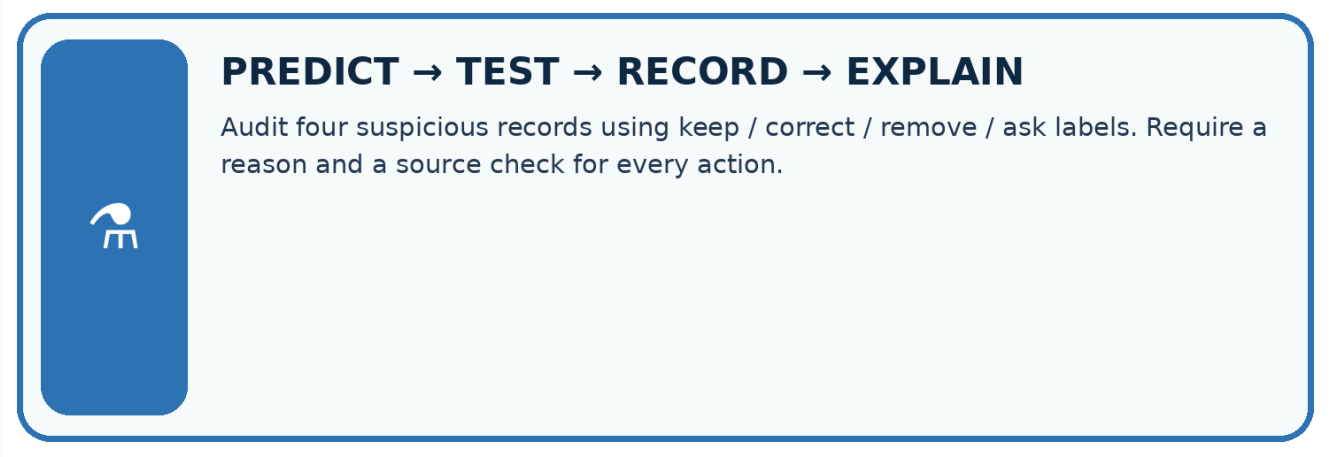
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Use the four supplied audit records. Choose keep, correct, exclude a confirmed duplicate from analysis, or ask for evidence; justify every action from the supplied source facts and preserve the raw record.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: compare summaries on a fresh five-value list

> Show sums, counts and sorted middle values before feedback.

The independent calculation

For 1, 2, 2, 3 and 12, calculate **mean** and **median**. Then replace 12 with 2 and recalculate. Explain the change.

Units and what-if status

For the selected independent task, treat the numbers as fictional durations in minutes. Replacing12 with2 is a requested what-if comparison, not evidence of a confirmed recording error. Explain the arithmetic change separately from permission to change an actual measurement.

Your first attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after the attempt

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Compare summaries independently

Week 17 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Page counts are 40, 42, 43 and 900. Should 900 be deleted immediately? Give two explanations to **investigate**.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Keep legitimate unusual books so the recommender can serve unusual readers.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

Page counts are 72,80,96,960. The last book may be an anthology. Should a cleaning script delete it automatically? Give a defensible investigation and document the outcome.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Keep a provenance record for unusual catalogue values

> A convenient cleaning rule is not evidence that a rare book is invalid.

This week’s concrete portfolio contribution

For the fictional recommender, inspect an unusual page count or processing time against edition, units and original source before changing it. Retain a verified long book, document a confirmed correction and avoid double-counting a proven duplicate of one event. A confirmed invalid record can be excluded from a declared analysis without erasing its raw source. Do not promise that five observations represent all readers or all library tasks.

Investigate unusual recommender features and show **evidence**-backed repairs instead of removing inconvenient records.

Keep the checkpoint, system and human choice

THIS WEEK'S CHECKPOINT Keep legitimate unusual books so the recommender can serve unusual readers.

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

> **Write or draw your reasoning and evidence.**

What should remain under **human control**?

> **Write or draw your reasoning and evidence.**

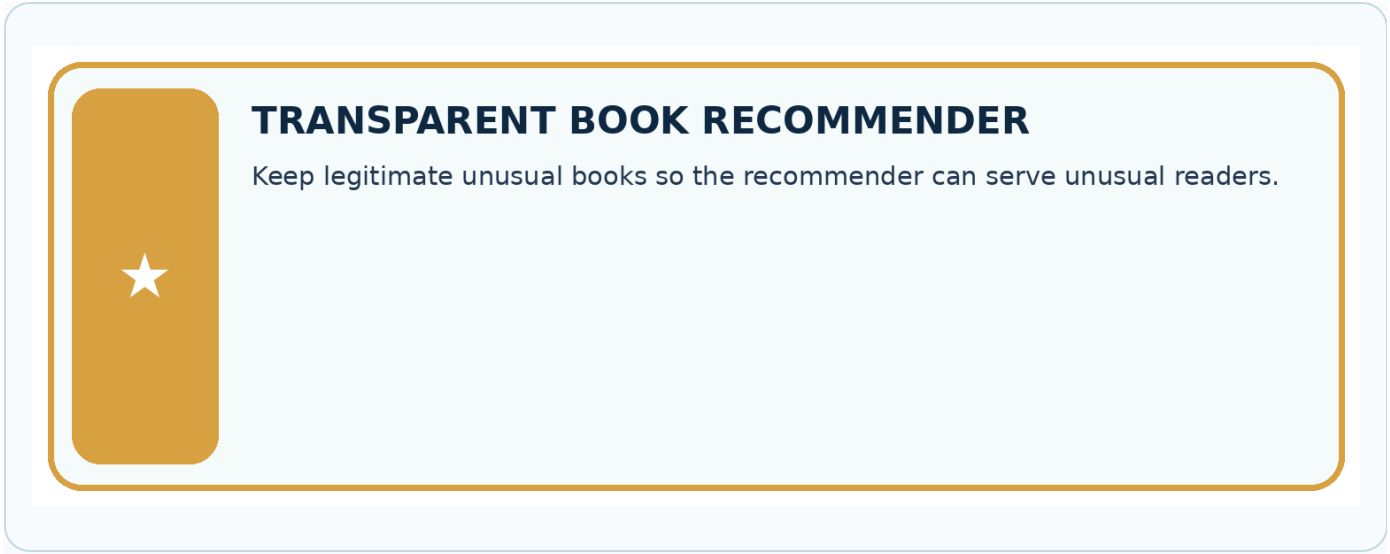
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

MAIN LESSON · SECTION 16

Finish: justify both the summary and any proposed repair

> Name the data, units, procedure and evidence limit.

Explain without notes

- Close your notes. Explain this goal in your own words: Calculate **mean** and **median**

My deciding step, evidence and one remaining question

Return on a changed case

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
noise	Variation or interference that obscures the information relevant to a task.	
outlier	An observation far from a typical pattern under a stated comparison; it is not automatically an error.	
duplicate	A repeated record or copy; copies are not independent new observations.	
investigate	Collect and examine evidence to answer a question rather than assuming a cause.	

Terms for the connected explanation

Word	Meaning
Mean	Sum of numeric values divided by their count
Median	The middle of sorted values or mean of the two middle values

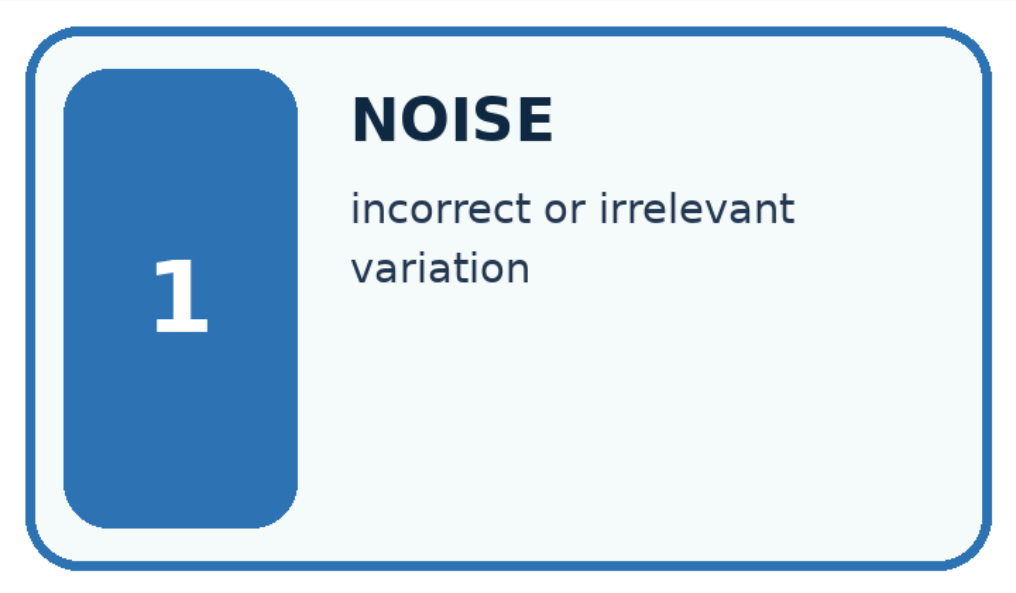
Word	Meaning
Unusual value	A value far from most others in a dataset
Variation	Differences among recorded values

▼ Earlier wording, alternatives and picture transcripts

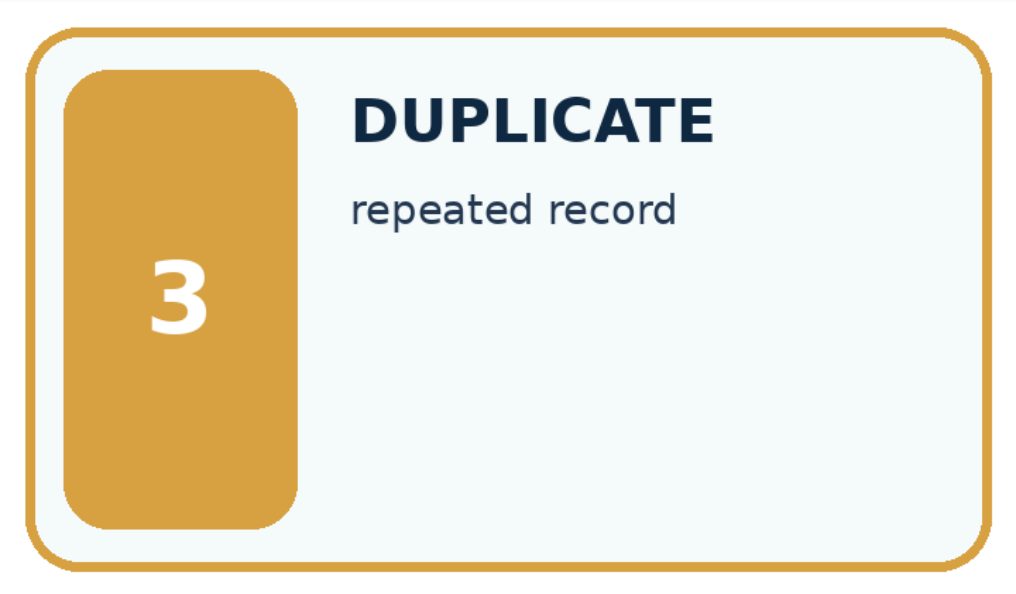
VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
OUTLIER
value far from most others


3
DUPLICATE
repeated record



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

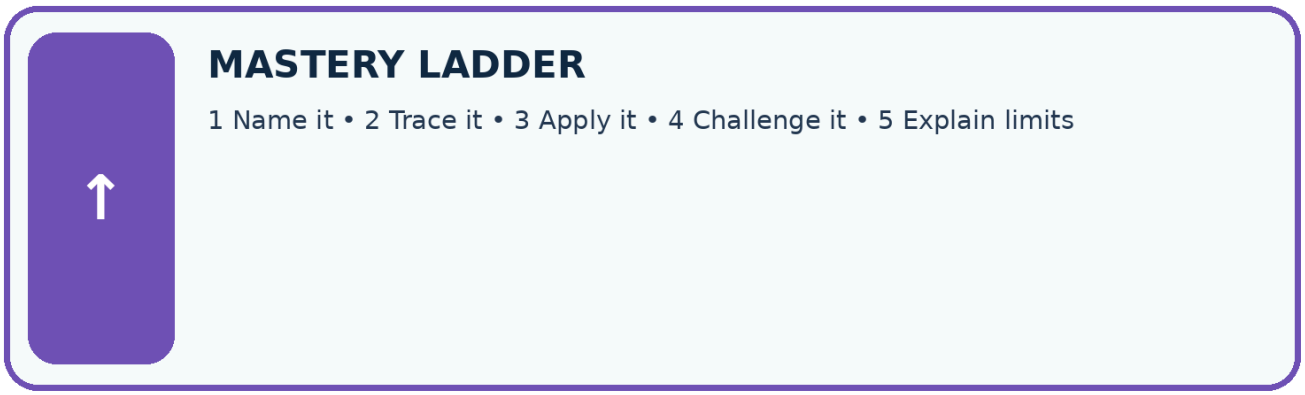
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Compare summaries independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

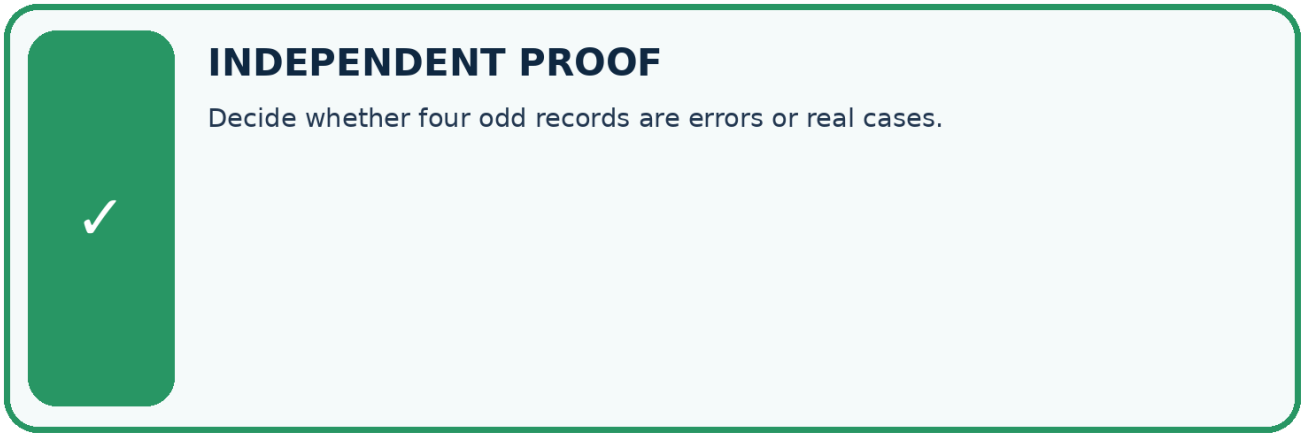
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ **Earlier wording, alternatives and picture transcripts**

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Compare summaries independently is the selected independent case.



EXPLANATION CHECK Count all observations, sort for the median, handle empty data explicitly and require source evidence for a repair.

Write or draw your reasoning and evidence.

▼ **Extra practice: An outlier needs investigation**

REFERENCE · SECTION 13

Extra practice: An outlier needs investigation

> Try the question before opening its worked explanation.

Read the supplied case and question

Recorded lengths are 90,95,100,105,900 pages.

Is 900 automatically wrong?

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

No. It is an **outlier**, but may be a real large book. Check catalogue, units and entry **process**.

Evidence or stage	Value or result
Book A	90
Book B	95
Book C	100
Book D	105
Book E	900

▼ Earlier wording, alternatives and picture transcripts

Week 17 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

An outlier needs investigation

Recorded lengths are 90,95,100,105,900 pages.

Is 900 automatically wrong?

No. It is an **outlier**, but may be a real large book. Check catalogue, units and entry **process**.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Calculate mean and median	Explain the deciding step, show a trace or test, and state one limit.
Investigate an unusual value before deleting it	Explain the deciding step, show a trace or test, and state one limit.
Explain how a summary can hide variation	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → **investigate** → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional library records five book-**process**ing times in minutes.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional library records five book-**process**ing times in minutes.

- Which labelled step matches this detail? Change a value only if **evidence** supports a correction; preserve the raw record.
- 1. **Median**
- 2. **Investigate** 18
- 3. Document repair
- 4. **Mean**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Document repair Change a value only if **evidence** supports a correction; preserve the raw record. The **mean** uses every value and is pulled upward by 18.

The **median** identifies the middle recorded time and stays at 3. Neither statistic declares the unusual **observation** invalid.

Deleting it simply because it changes the **mean** would hide potentially real slow work and change the question being answered.

Changed-case question

- Changed-case practice: An authorised log confirms 18 was a transcription error for 3. Which conclusion is supported by the supplied facts?
- 1. Corrected **mean** is 3 minutes and **median** is 3.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: Corrected **mean** is 3 minutes and **median** is 3. Values become 2,3,3,4,3 with sum 15; sorted middle remains 3.

Repair and save your work

Review Over**fitting** Intuition (ai-2/16). Goal: Compare **training** and validation **evidence** for model capacity. Short practice: Does zero **training** error justify choosing Complex in the original record?.

Return to this week and explain what you changed.

Review Decision Trees (ai-2/14). Goal: Fit a root question and leaf **prediction**s from **training** examples. Short practice: Why does changing colour alone not change this tree's **prediction**?.

Return to this week and explain what you changed.

Review **Training** and Testing (ai-2/11). Goal: Choose a split unit that matches the intended **prediction** claim. Short practice: How many distinct books are in each repaired role, and why report that as well as scan counts?.

Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

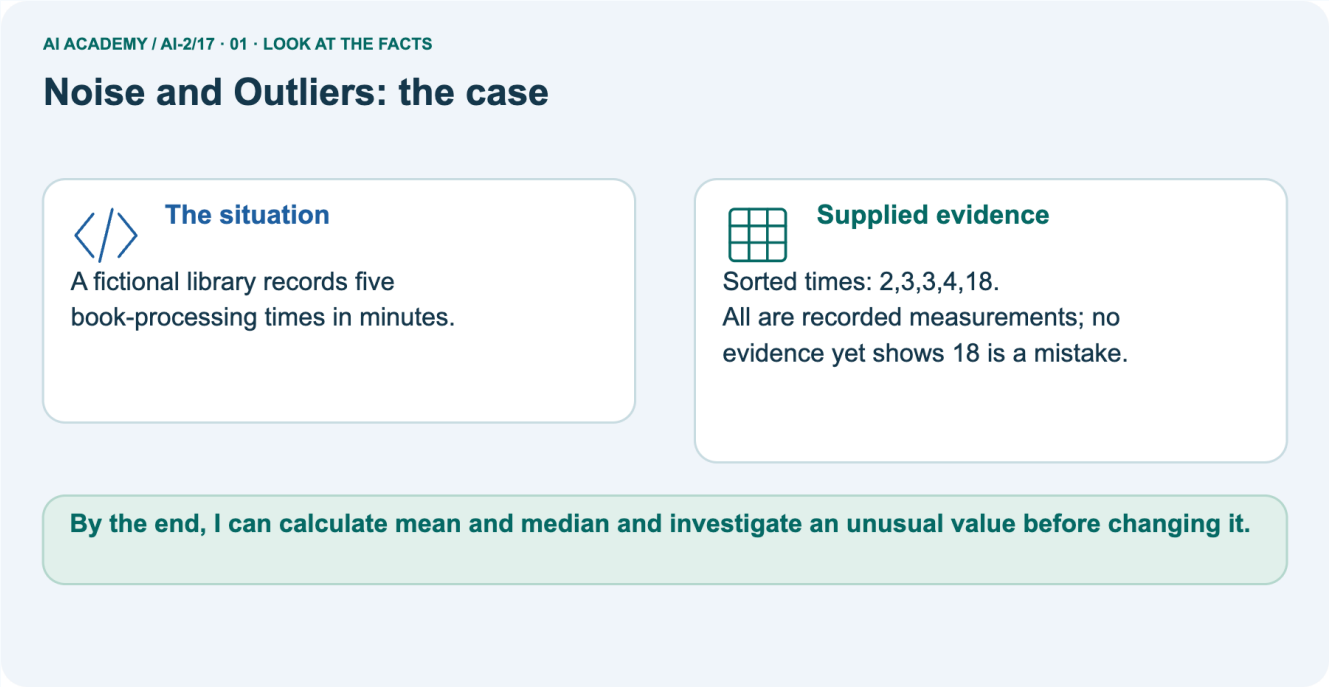
▼ Reference: the six-picture journey

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Noise and Outliers: the case

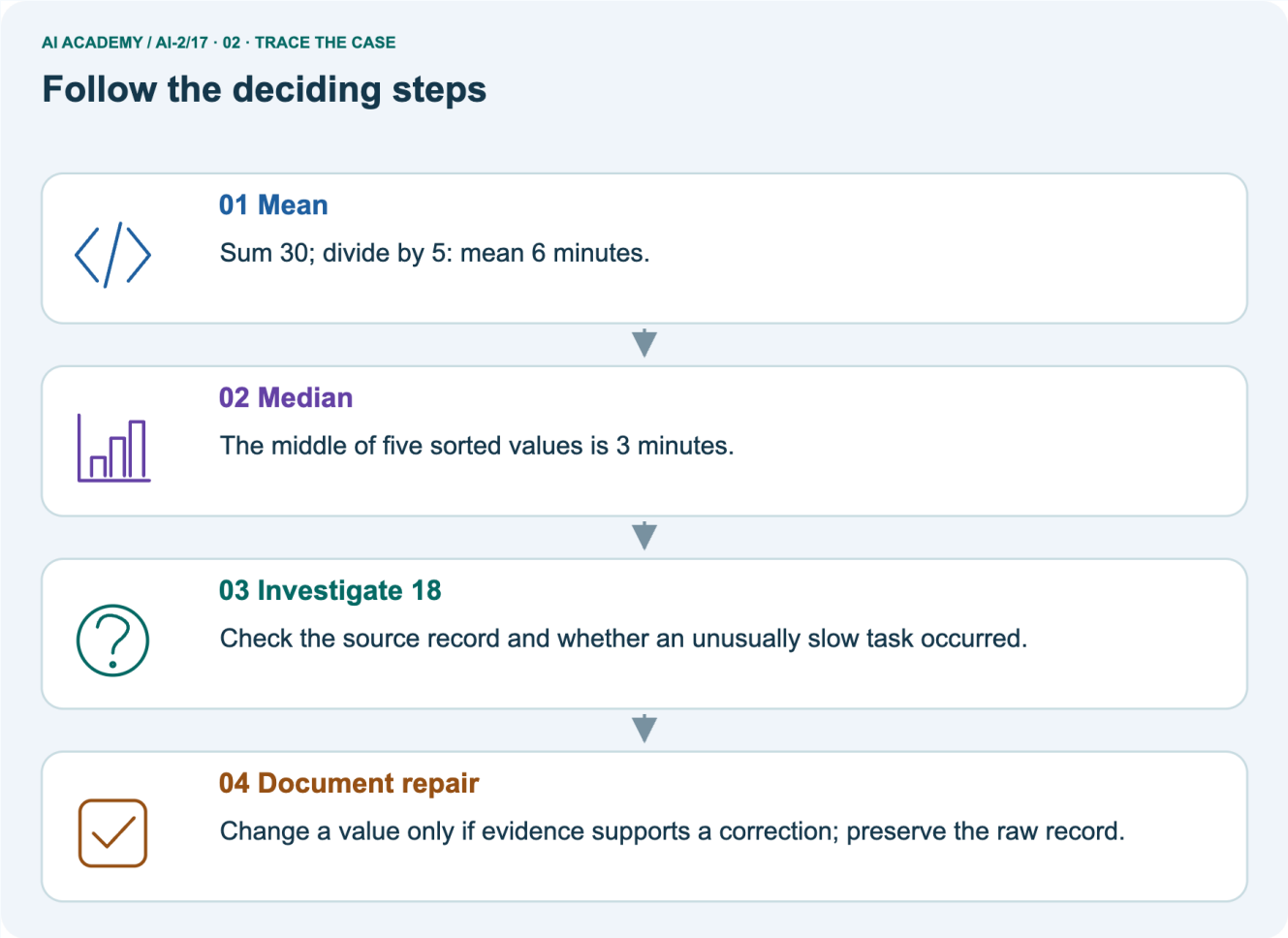


Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional library records five book-**process**ing times in minutes. Supplied facts: Sorted times: 2,3,3,4,18. All are recorded measurements; no **evidence** yet shows 18 is a mistake.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

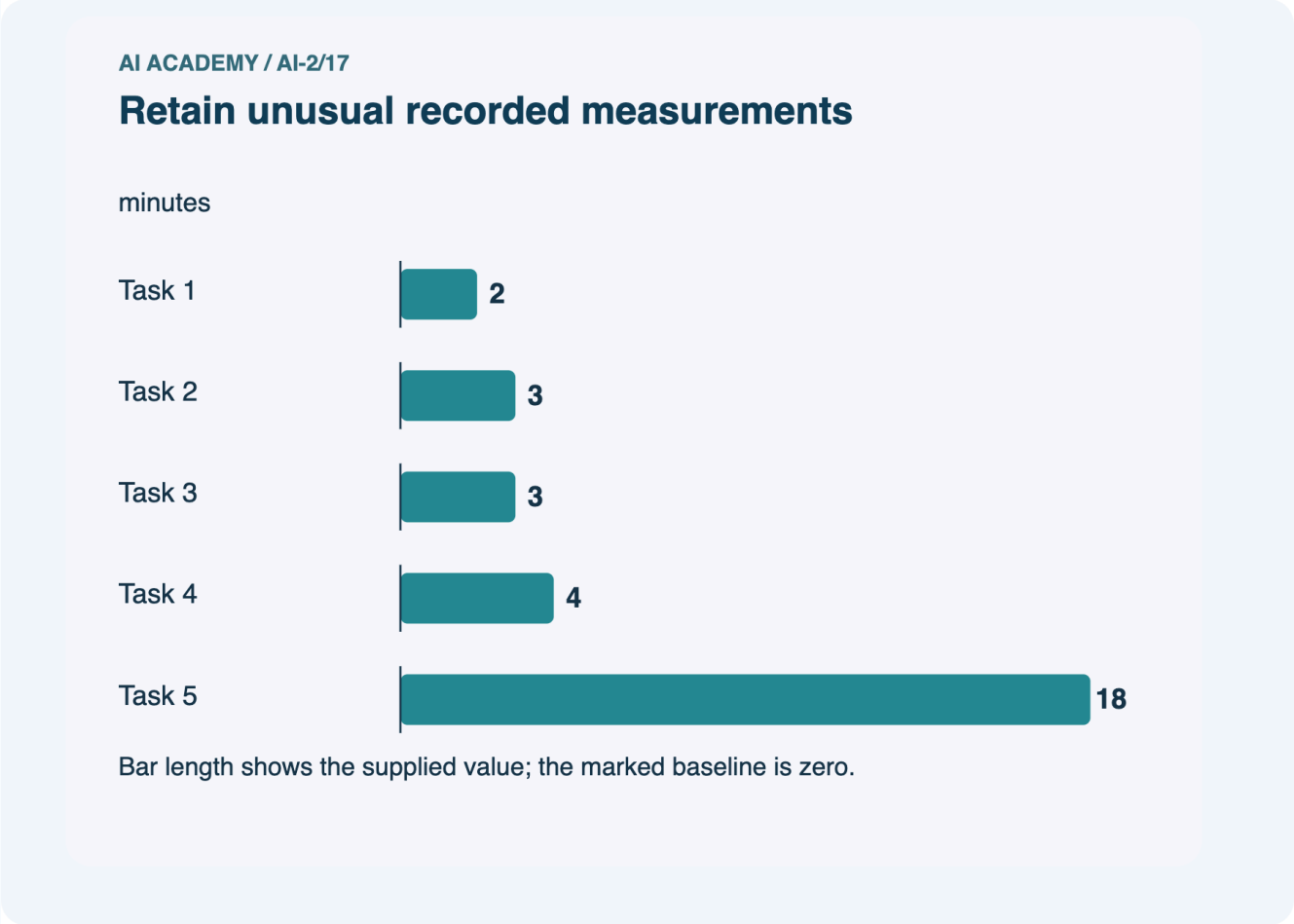


Follow the labelled picture with the matching explanation. Open the vector diagram

Mean: Sum 30; divide by 5: **mean** 6 minutes. → **Median**: The middle of five sorted values is 3 minutes. → **Investigate** 18: Check the source record and whether an unusually slow task occurred. → Document repair: Change a value only if **evidence** supports a correction; preserve the raw record.

Point to each numbered stage and explain which recorded **input** it uses.

03. Retain unusual recorded measurements

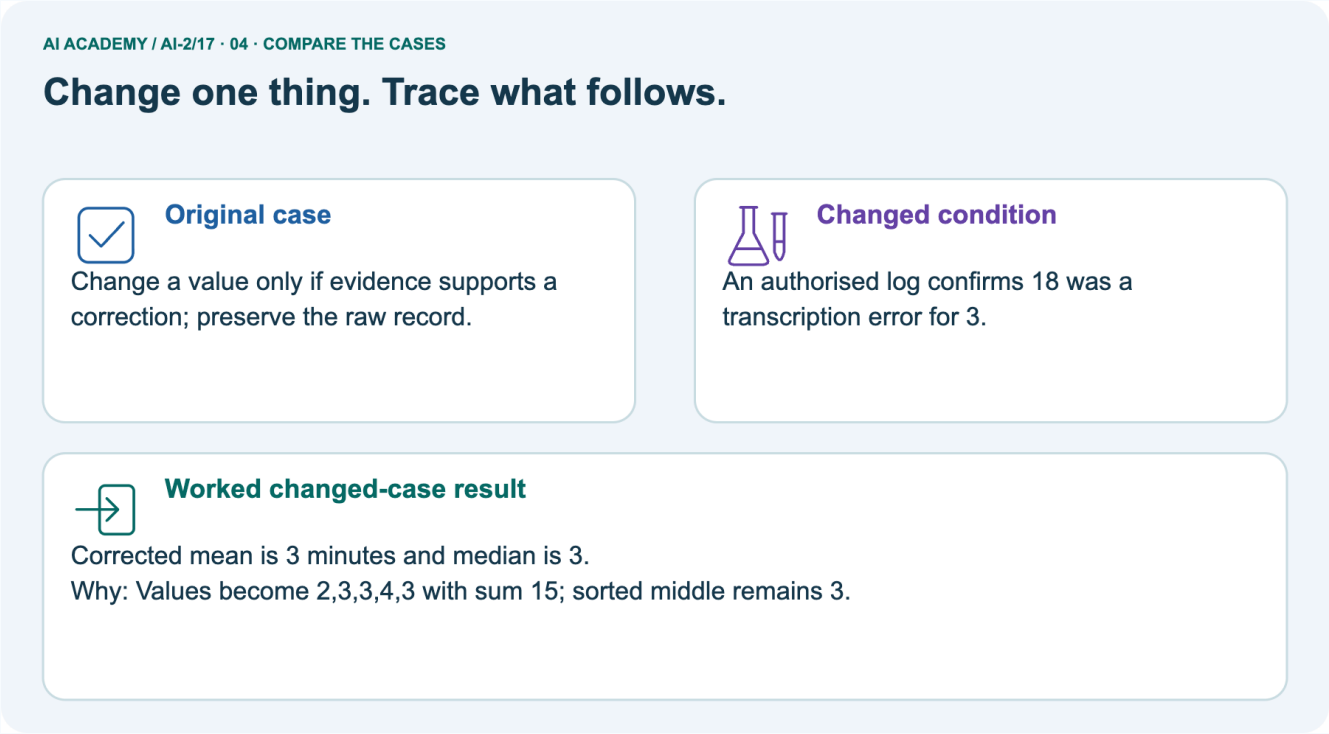


Follow the labelled picture with the matching explanation. Open the vector diagram

Retain unusual recorded measurements. Units: minutes. Task 1: 2. Task 2: 3. Task 3: 3. Task 4: 4. Task 5: 18.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



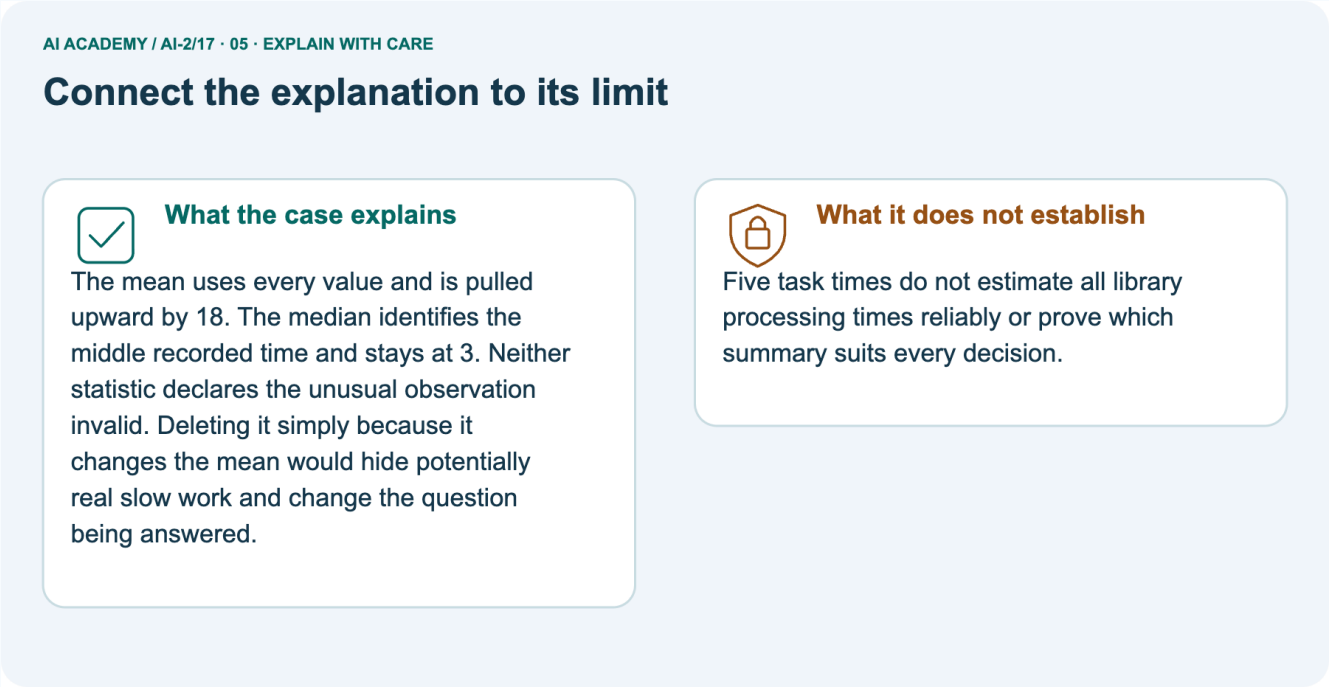
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Change a value only if **evidence** supports a correction; preserve the raw record. Changed condition: An authorised log confirms 18 was a transcription error for 3.

Expected result: Corrected **mean** is 3 minutes and **median** is 3. Reason: Values become 2,3,3,4,3 with sum 15; sorted middle remains 3.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

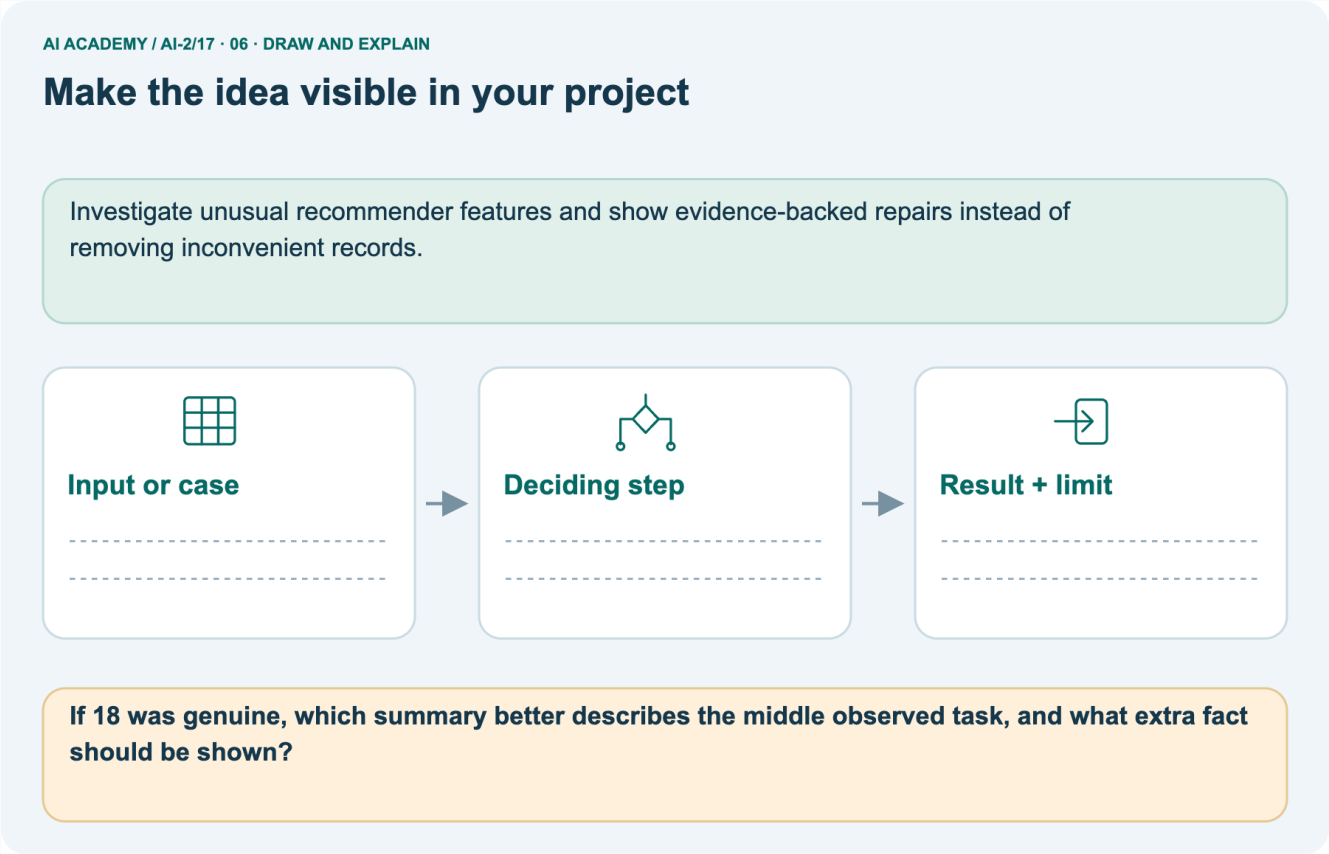
The **mean** uses every value and is pulled upward by 18. The **median** identifies the middle recorded time and stays at 3. Neither statistic declares the unusual **observation** invalid.

Deleting it simply because it changes the **mean** would hide potentially real slow work and change the question being answered.

Limit: Five task times do not estimate all library **process**ing times reliably or prove which summary suits every decision.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Investigate unusual recommender features and show **evidence**-backed repairs instead of removing inconvenient records. Draw your **input**, deciding step, result and limit.

Supported question: If 18 was genuine, which summary better describes the middle observed task, and what extra fact should be shown?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Noise and Outliers: the case

02. Follow the deciding steps

03. Retain unusual recorded measurements

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.